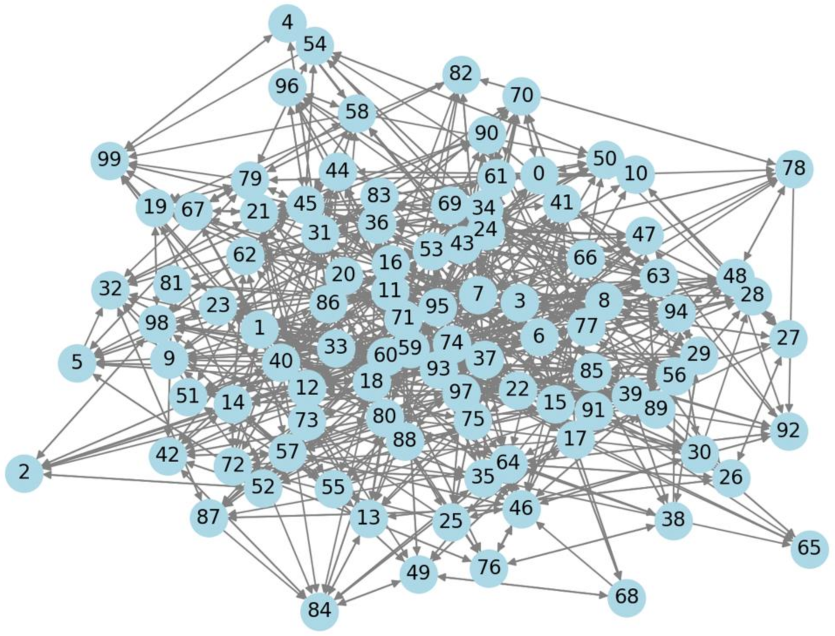
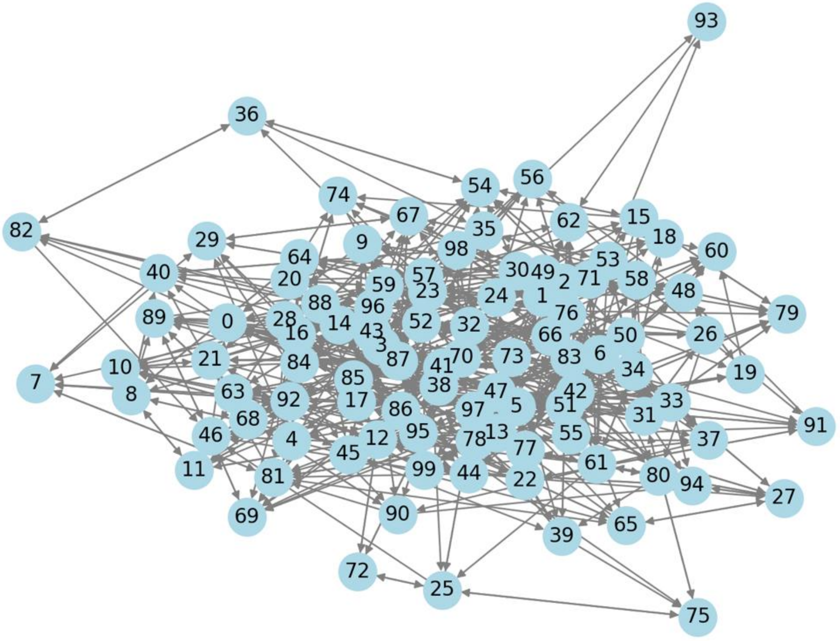
S12: Network Diagnosis

Why to learn network-level metrics?

- How to understand the structure of and compare large networks (with many nodes)?





Network diagnosis

Business analyst = a clinical doctor

You cannot always do a surgery for diagnosis of a complicated disease

Characterize a network = run lab tests on relevant biometrics

Metrics = Biometric results

From the metrics, we can depict the diseases/networks' problems

Characterize the network

- Cohesion
  - Capture the **connectedness** among nodes.
  - It measures how **tightly bound** the nodes are and how easily they can communicate with each other.
  - A **highly cohesive network** has strong internal connections
  - A **low-cohesion network** is loosely connected and may have isolated groups.

- Other characteristics
- E.g., centralization
  - Whether the network is centered on one or a few important nodes
  - **Dispersion** concept

Average degree

- The average of node-level degree
- How many friends to have on average

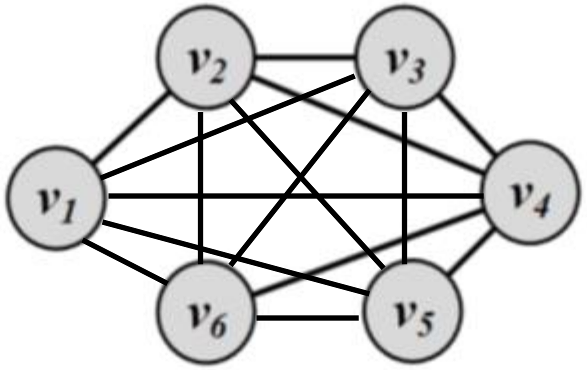
Some fun facts

- HS Friendships (CJP 09)
- 6.5
- Romances (BMS 03)
- 0.8
- Facebook (Marlow 09)
- 120

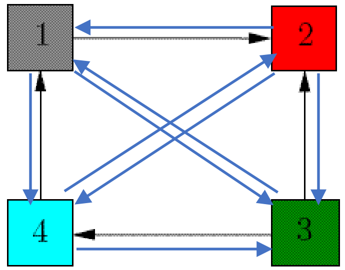
- Co-authors (Newman 01, GLM 06)
- Bio
- 15.5
- Econ
- 1.7
- Math
- 3.9
- Physics
- 9.3

Density

- Network density:
- A network's *density* = |edges (arrows)| / |Edges in its fully connected network|
- Fully connected network: there is an edge between any pair of nodes (an arrow from any node to any other nodes)
- |Edges in a fully connected network|
- = total number of possible edges between all pairs of nodes
- =*n(n-1)/2*, where *n* is the number of nodes for an undirected network
- = *n(n-1)* for a directed network



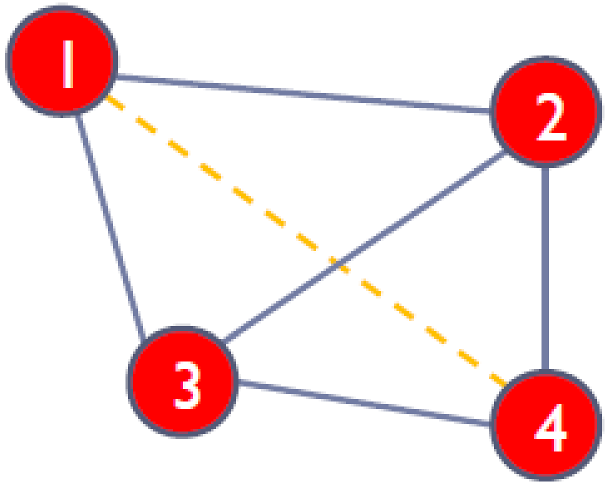
	v_1	v_2	v_3	v_4	v_5	v_6
v_1	0	1	0	0	0	0
v_2	1	0	1	1	0	0
v_3	0	1	0	1	0	0
v_4	0	1	1	0	1	1
v_5	0	0	0	1	0	0
v_6	0	0	0	1	0	0



$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Example: undirected

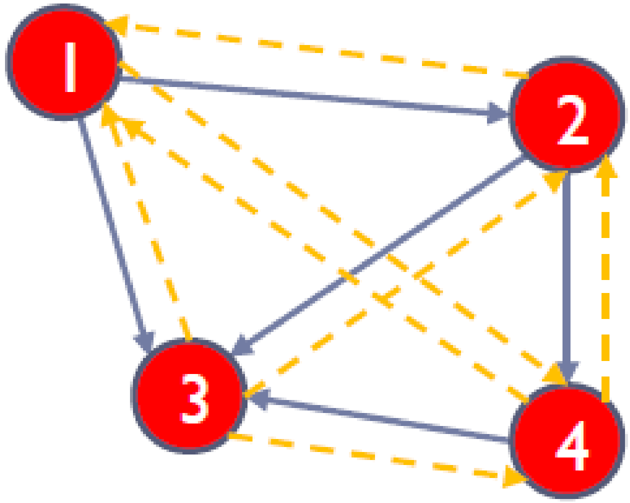
- In the example network to the right density=5/6=0.83 (i.e. it is a fairly *dense* network; opposite would be a *sparse* network)
- It is a common measure of how well connected a network is (in other words, how closely knit it is)



density = 5/6 = 0.83

Example: directed

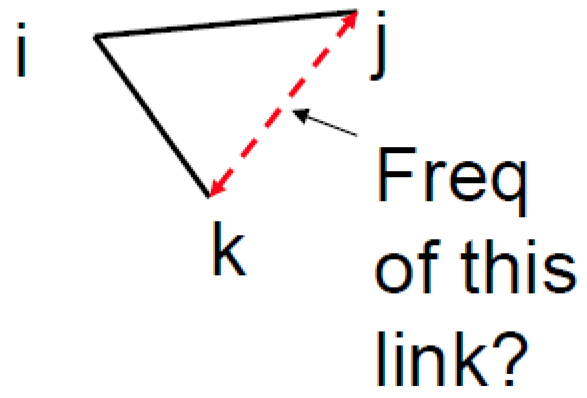
- A directed graph will have half the density of its undirected equivalent, because there are twice as many possible edges, i.e., n(n-1)



density = 5/12 = 0.42

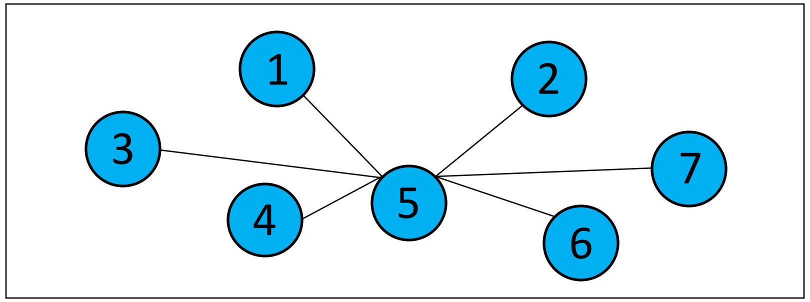
Network clustering (transitivity)

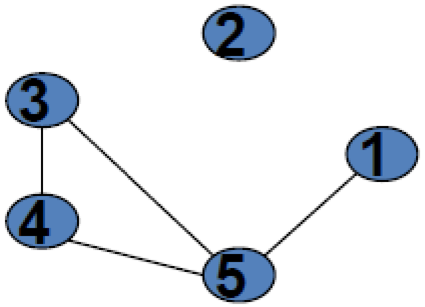
- Node level: What fraction of my friends are friends of each other?
- Average clustering
- Network clustering: “Density” of clustering
- “Density” of triangles (closed triplets)

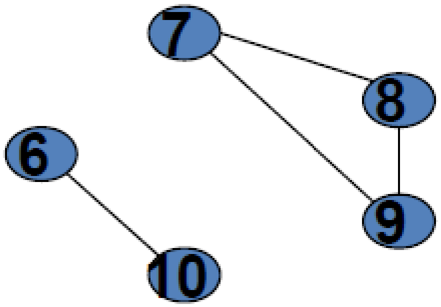


Calculation

- The number of **closed** triplets (triangles) in the node's neighborhood over the number of triplets (connected 3 nodes) in the neighborhood.
- 0 triangles -> 0 clustering coefficient
- 2 triangles, 4 triplets -> 0.5 clustering coefficients





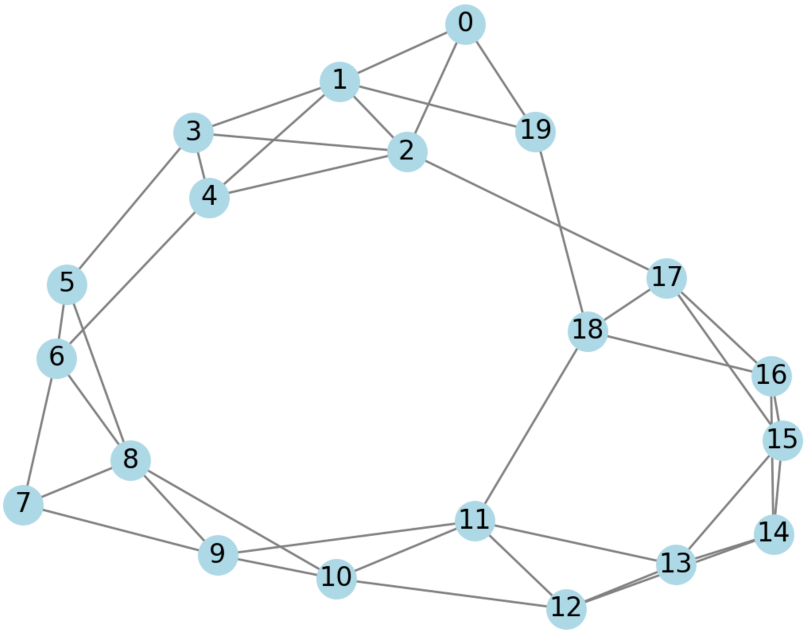


Clustering vs. density

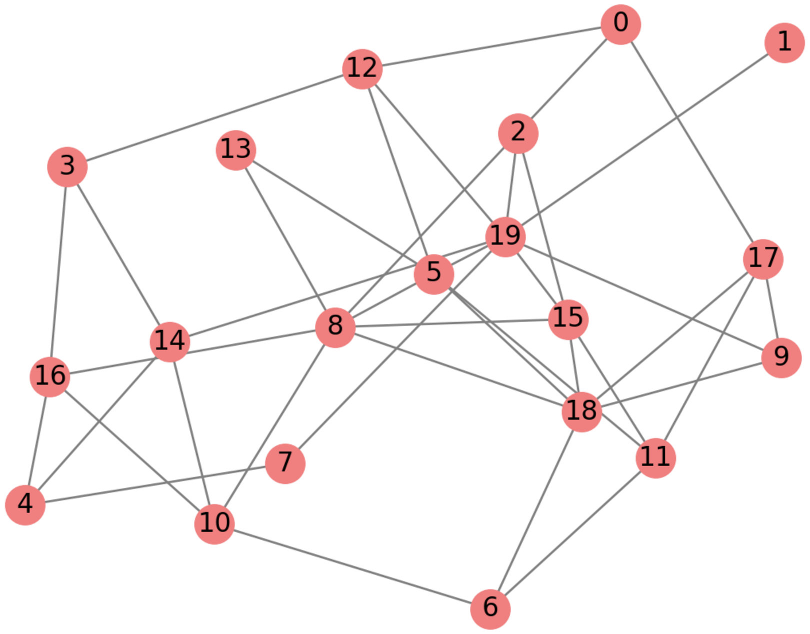
- Just like density, clustering captures how tightly-knit/**sparce** the network is.
- Same level of density
- Clusters of edges → local groups → tightly-knit communities

- Evenly distributed edges
- More **dispersed (evenly distributed edges)**

Graph with High Clustering (0.4467)

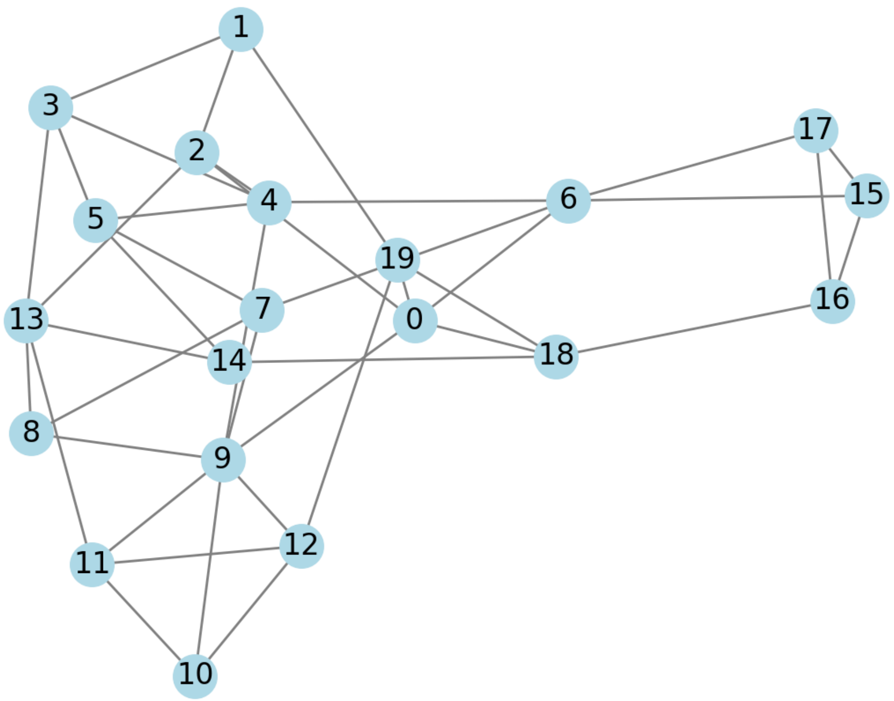


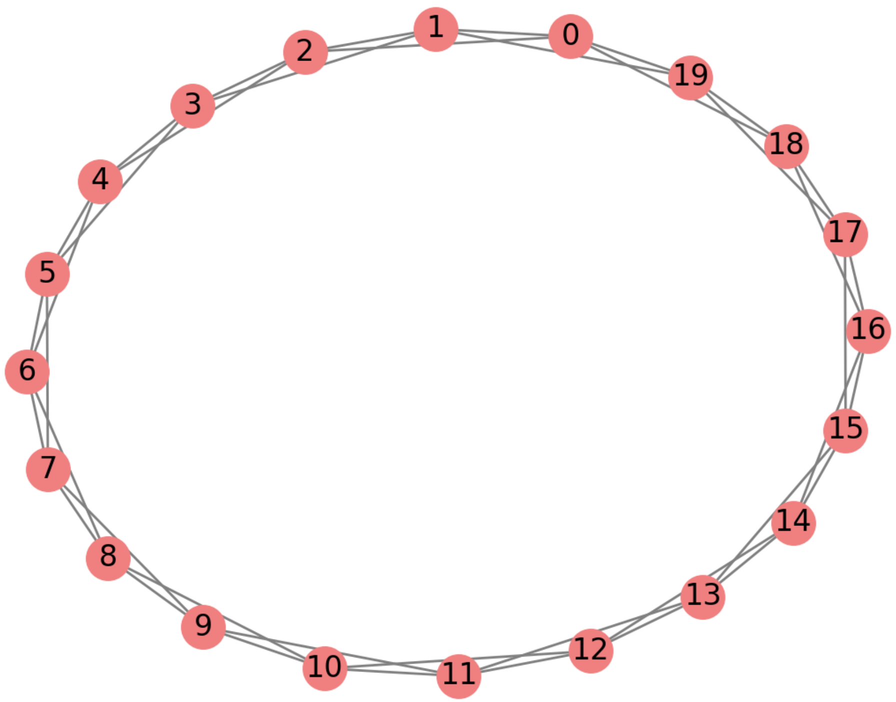
Graph with Low Clustering (0.1671)



Transitivity ?= how fast to to spread out information

- The same clustering, but
  - Do they take the same time to reach a node on average?
  - Which network has fast information spreading?





Density ?= how fast to to spread out information

- The same density, but
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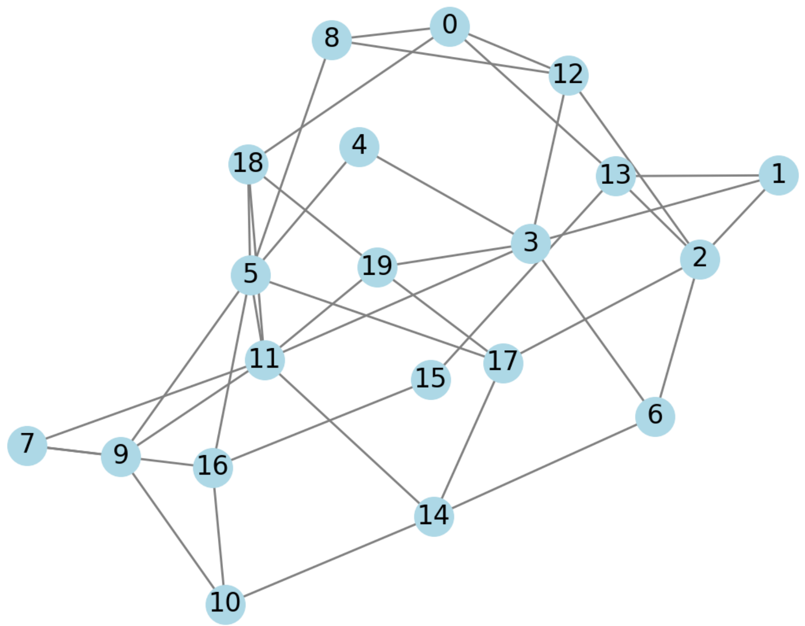
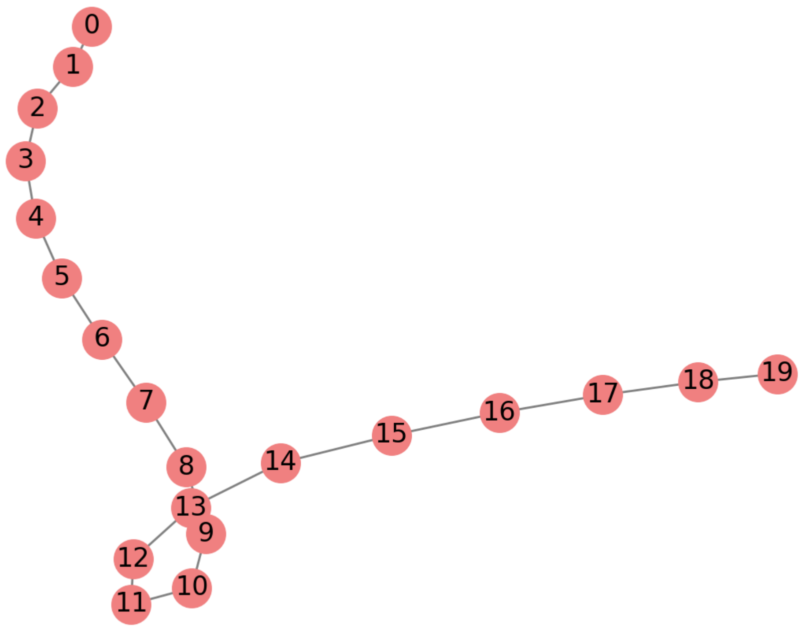
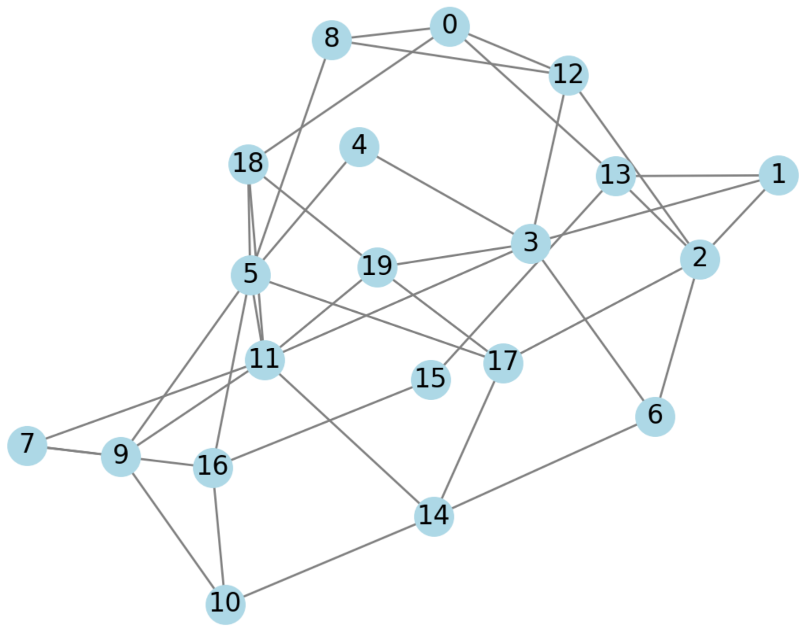


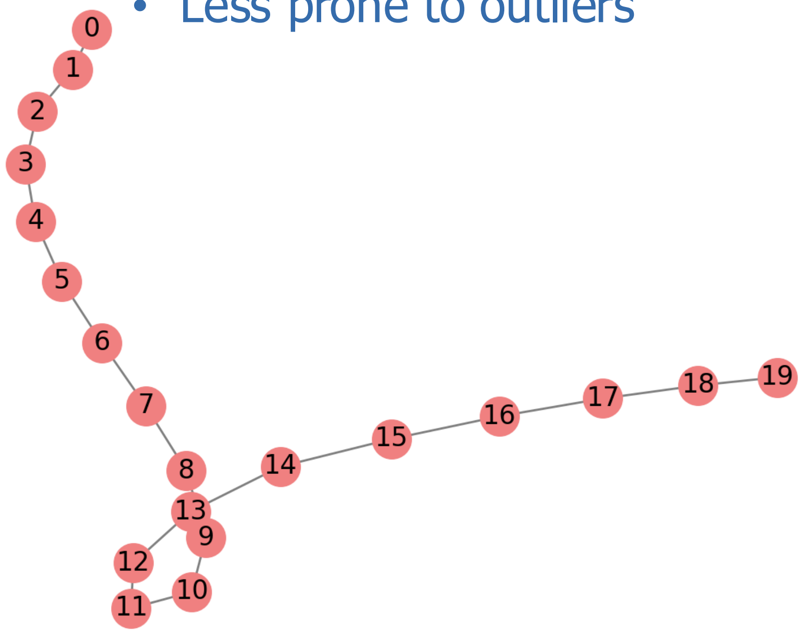


Diameter

- How close are nodes to each other:
- How long does it take to reach average node?
- How fast will information spread?
- Defined on undirected networks only



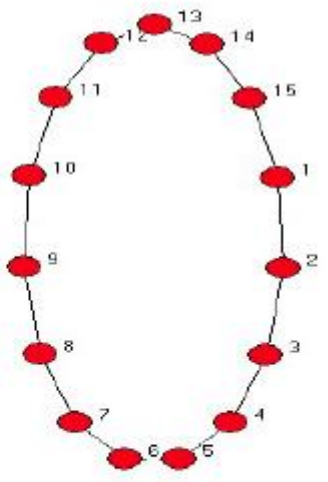
- Maximum Geodesic Distance
  - Longest shortest path
  - If unconnected, of largest component
- Average Geodesic Distance
  - Less prone to outliers

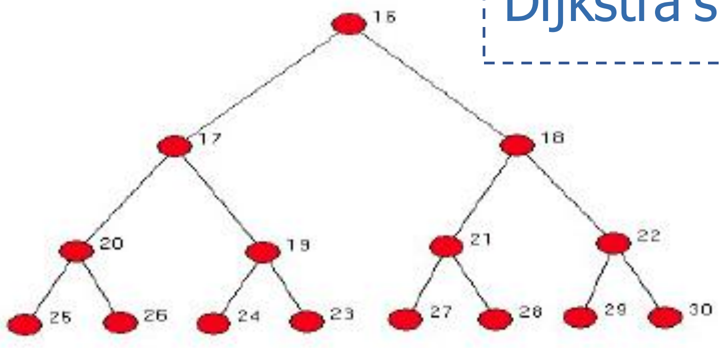


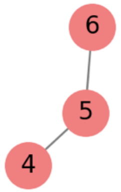
What are the diameters?

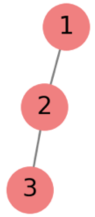
K-Level has a diameter of 2K

Dijkstra's algorithm









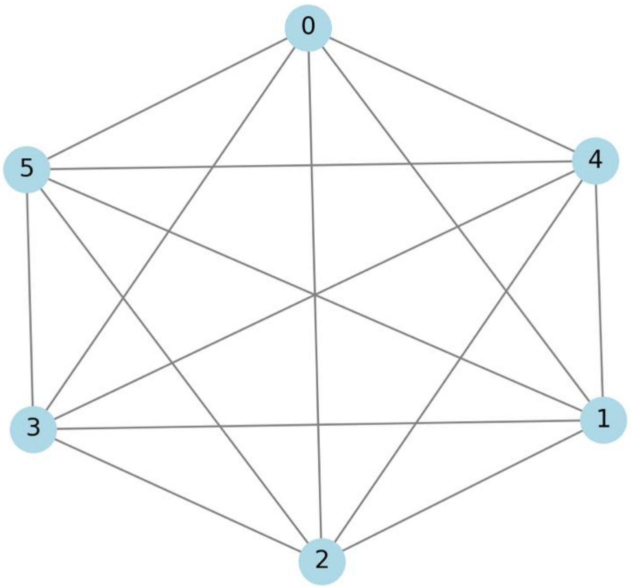
Some fun fact

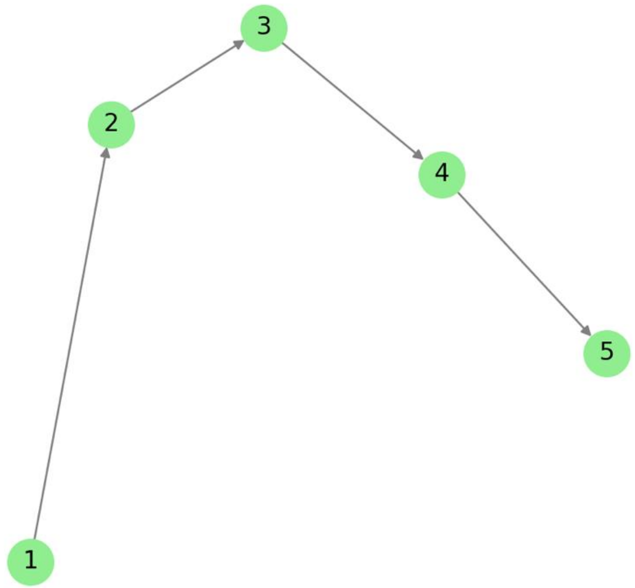
- Co-Authorship studies
  - Grossman (2002) Math mean 7.6, max 27,
  - Newman (2001) Physics mean 5.9, max 20
  - Goyal et al (2004) Economics mean 9.5, max 29
- Facebook
  - Backstrom et al (2012) – mean 4.74 (721 million users)

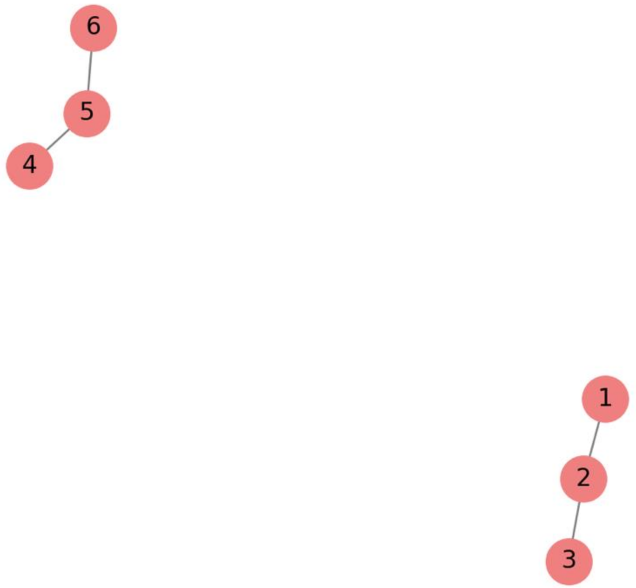
Connectivity

- **Definition:** The minimum number of nodes or edges that need to be removed to disconnect the network.
- **Types:**
  - **Node Connectivity:** Minimum number of nodes that, if removed, would disconnect the graph.
  - **Edge Connectivity:** Minimum number of edges that, if removed, would disconnect the graph.
- **Interpretation:**
  - High connectivity implies robustness and redundancy.
  - Low connectivity indicates vulnerability to disconnection.

Practice

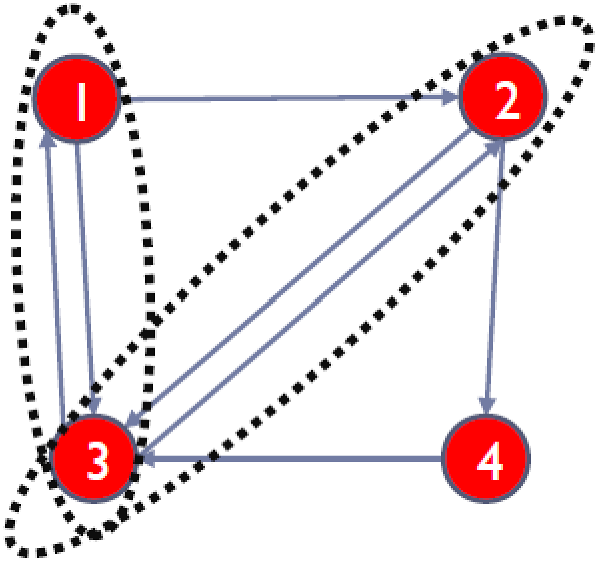
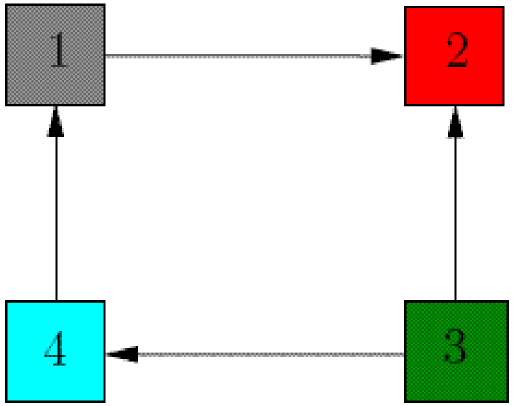






Max-Flow Min-Cut Algorithm

Python

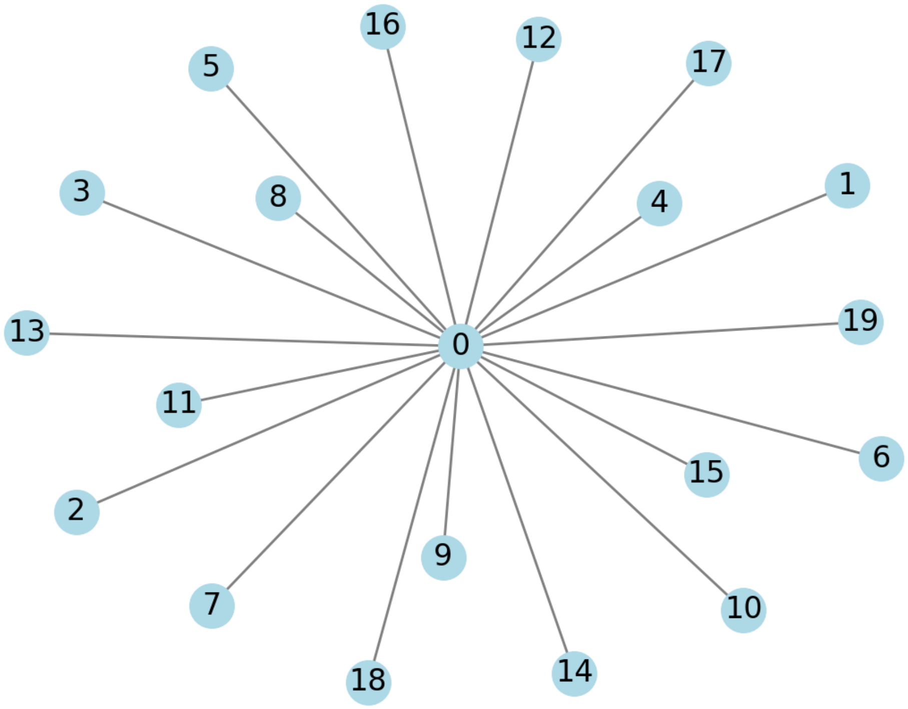
- ```
  avg_in_degree = sum(dict(G.in_degree()).values()) / num_nodes
  avg_out_degree = sum(dict(G.out_degree()).values()) / num_nodes
  avg_degree = sum(dict(G.degree()).values()) / num_nodes
  ```
- ```
  density = nx.density(G)
  ```
- ```
  network_clustering = nx.transitivity(G)
  ```
- ```
  diameter = nx.diameter(G)
  ```
- ```
  node_connectivity = nx.node_connectivity(G)
  edge_connectivity = nx.edge_connectivity(G)
  ```
  ```

Reciprocity (directed network only)

- The ratio of the number of related (connected) nodes which are reciprocated (i.e., two-way edges between the nodes) over the total number of related (connected) nodes (one-way or two-way) in the network
- A useful indicator of the degree of mutuality and reciprocal exchange in a network, which relate to social cohesion
- Only makes sense in directed graphs

Examples

- In the example to the right this would be 2/5=0.4 (whether this is considered high or low depends on the context)

- In the second example, this would be?



Reciprocity for network = 0.4



Compute reciprocity

- Reciprocity=|Symmetric edges|*0.5/(|asymmetric edges|+|Symmetric edges|*0.5)
- Symmetric edges: M(x,y)=M(y,x)=1
- Asymmetric edges: M(x,y)+M(y,x)=1
- |Symmetric edges|+|Asymmetric edges|=|edges|

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Python

- ```
  # Reciprocity
  reciprocal_edges = sum(1 for u, v in G.edges() if G.has_edge(v, u))
  reciprocity = reciprocal_edges / (num_edges-reciprocal_edges)
  print(reciprocity)
  ```
  ```

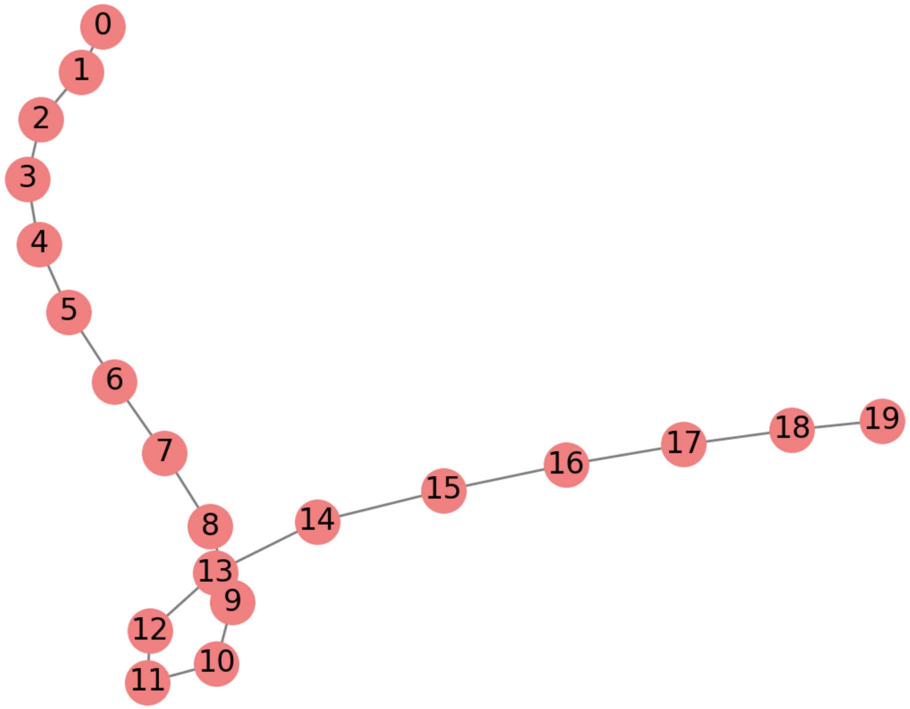
Centralization

- Characterizes the amount to which the network is centered on one or a few important nodes
- Centralized networks have a small amount of central nodes and many non-central nodes
- Decentralized networks have little variation b/w the centrality each vertex possesses

Centralization Type	**Measures**	**Key Question**
Degree Centralization	Node Degree	Is the network controlled by a few well-connected nodes?
Betweenness Centralization	Shortest Paths	Do a few nodes act as key brokers?
Closeness Centralization	Distance to Others	Are some nodes much closer to all others?
Eigenvector Centralization	Influence	Do a few nodes dominate in influence?

Visualization





Computation

Centralization is calculated as:

$$C = \frac{\sum_{i=1}^{N} (\max(C_i) - C_i)}{(N-1)(N-2)}$$

where:

- C_i is the **centrality score** of node i,
- $\max(C_i)$ is the **highest centrality score** in the network,
- N is the **number of nodes.**

- The denominator represents the maximum possible degree centralization for that network size
- If every node has the same degree, what will be the centralization?

Python

```
max_degree = max(degrees.values())
N = len(G.nodes())
centralization = sum(max_degree - d for d in degrees.values()) / ((N - 1) * (N - 2))

print(centralization)
```

Review: Characterize a network

- Average degree
- Density
- Clustering
- Diameter
- Connectivity
- Centralization

- The portion of closely knit communities
- How fast to reach other nodes from a node
- How many nodes to reach immediately from a node on average
- How well connected for the entire network
- How robust a network is to disconnection
- Are there any “super” nodes that can control the majority of the network?

MGP 5b

- Your team provides consulting solutions for an election campaign on social media. The campaign will disseminate promotional posts to certain number of users (seeding) and let it evolve within the network.
- There are two candidate social media platforms (networks can be imported using the code below), and due to the budget constraint, you can only afford to run the campaign on one.
- Your client needs your advices based on different what-if settings and goals.
- In specific, please answer:

```
import networkx as nx

G1 = nx.read_edgelist("graph1.edgelist", nodetype=int)
G2 = nx.read_edgelist("graph2.edgelist", nodetype=int)
#Please put the two files under your pycharm project folder first
```

MGP 5b

- If the goal is to reach more people and the post is highly (near a probability of 1) and constantly (across different person) contagious in triggering reposting,
  - Which network to pick if with unlimited time and a moderate number of random seeds (e.g., you can let 10 randomly chosen persons initialize the posts)?
  - Which network to pick if with moderately limited (not too short) time and a moderate number of random seeds?
  - Which network to pick if with very limited time and a moderate number of random seeds?
- For the same goal, if the post is not contagious in triggering reposting and there can be only 1 seed that is up to your pick, which network to pick?

MGP 5a

- For the same goal, if the post is only contagious in very well-connected communities (i.e., subgroups with more than 3 persons where everyone knows every other), which network to pick?
- If the goal is to reach all nodes in shorter time and the post is highly (near a probability of 1) and constantly (across different person) contagious in triggering reposting, which network to pick?


MELIORA